INCH-POUND

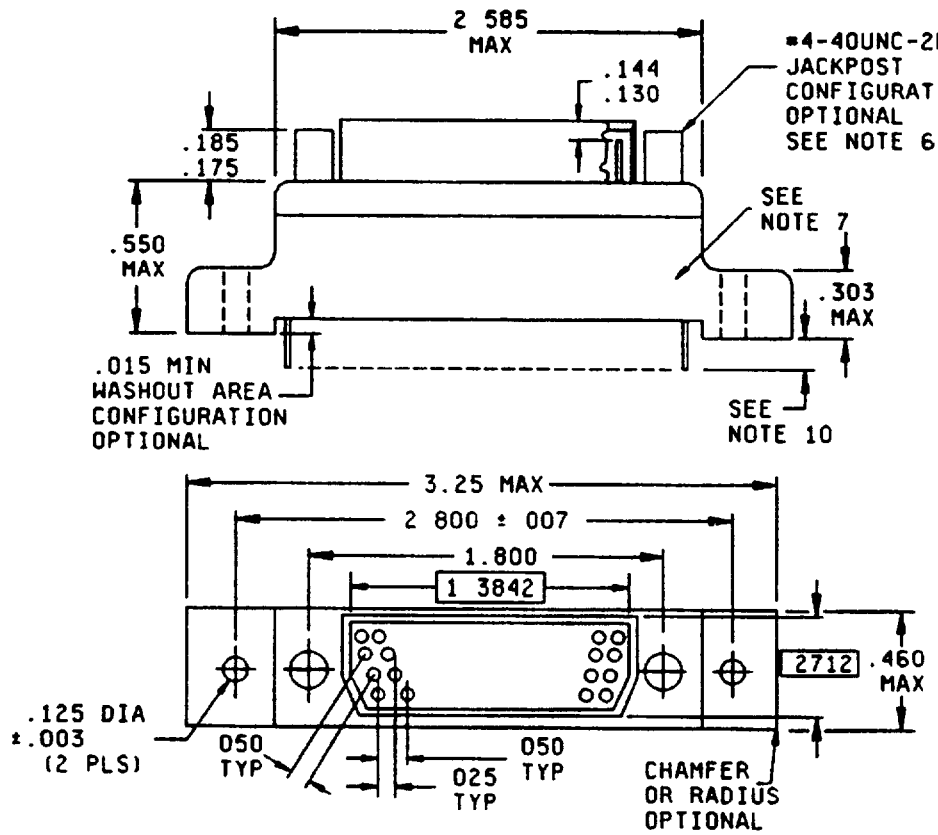
MIL-C-83513/27A
14 Oct 1993
SUPERSEDING
MIL-C-83513/27
11 February 1992

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, RECEPTACLE, MICROMINIATURE, POLARIZED SHELL, STRAIGHT, SOCKET CONTACTS, 4 ROW, SOLDER TYPE, 100 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83513

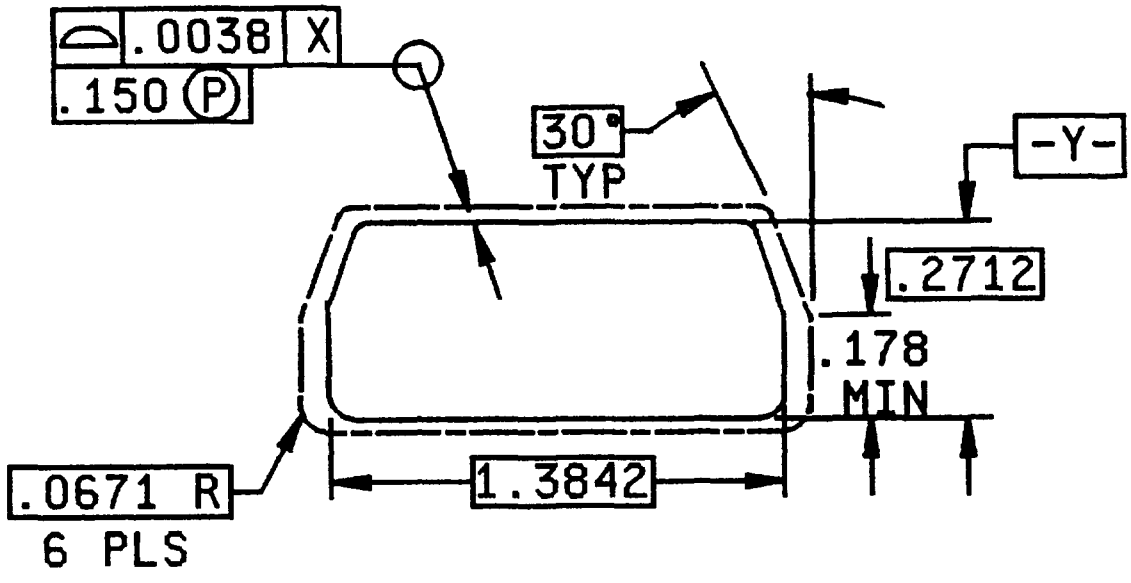


Inches	mm
.003	0 08
.007	0.18
.0038	0 097
.0671	1.704
.015	0.38
.025	0.64
.050	1.27
.125	3.18
.130	3.30
.144	3.66
.150	3.81
.165	4.19
.175	4.44
.185	4.70
.2712	6.888
.303	7.70
.325	8.26
.460	11.68
.550	13.97
1.3842	35.159
1.452	36 88
1.800	45.72
2.585	65.66
2.800	71.12

FIGURE 1. Connector, receptacle, .050 spacing

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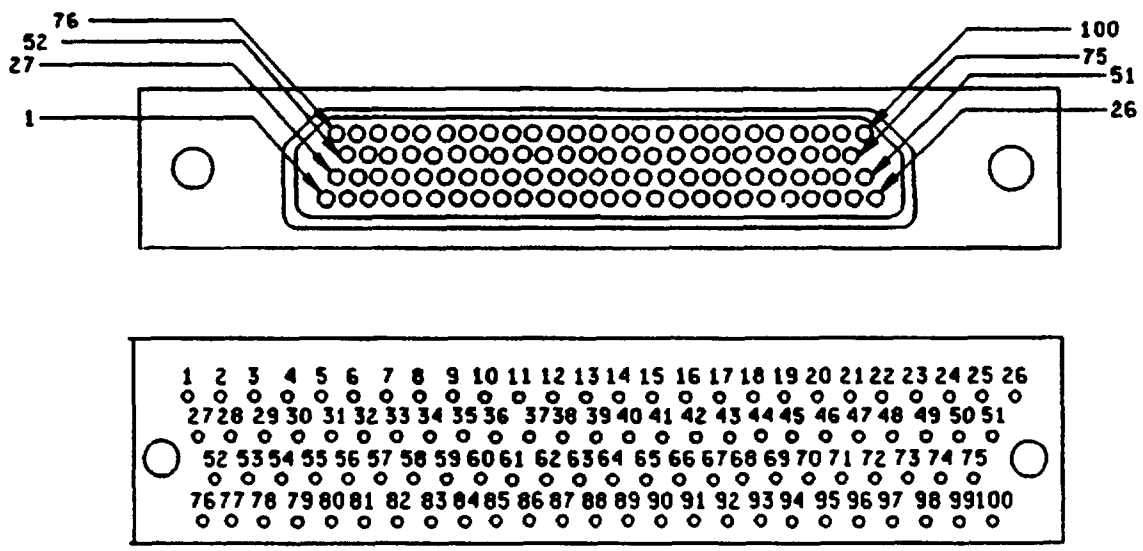
DETAIL OF INTERFACE

NOTES

1. Dimensions are in inches.
2. Metric equivalents are given for information only
3. Unless otherwise specified, tolerances are ±.005 (0 13 mm).
4. Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513. Plastic molding shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
5. Metal shell shall be of material in accordance with MIL-C-83513 for class M parts
6. Jackpost (permanently attached), when specified. Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35.
7. Separately molded plastic body (if used) shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
8. Wire termination sockets shall conform to QQ-W-343, no 24 AWG copper.
9. Interfacial seals are included with each type receptacle connector
10. Termination lengths available .109 (2 77 mm), 140 (3.56 mm), or .172 (4 37 mm) The tolerance shall be ±.015 (0.38 mm) for all termination lengths.

FIGURE 1 Connector, receptacle, 050 spacing - Continued

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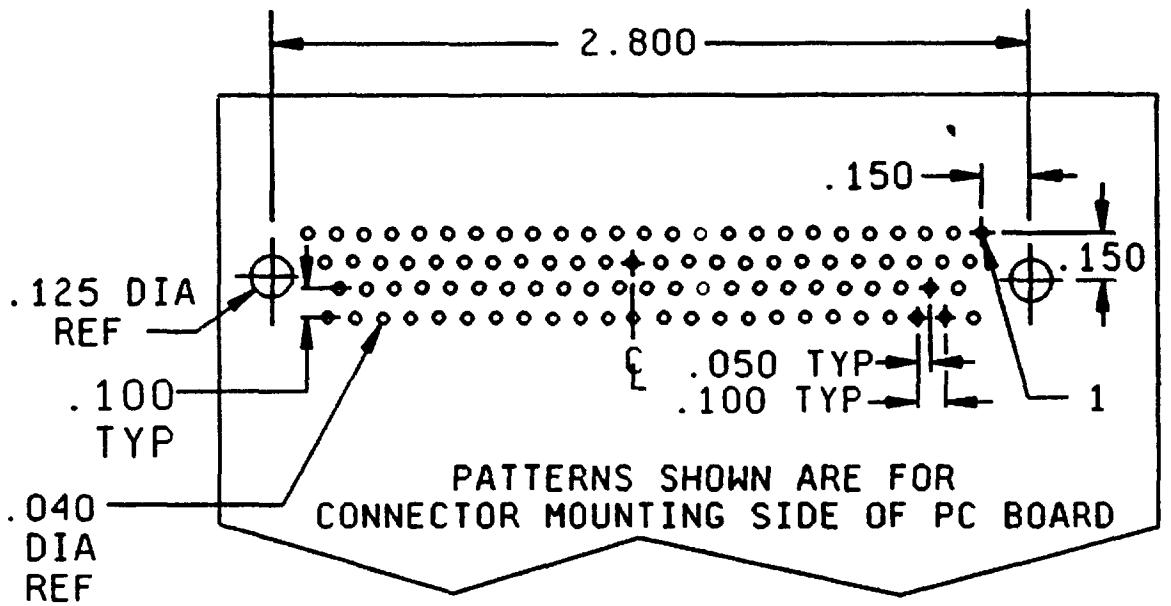
TERMINATION VIEW - VERTICAL MOUNT

NOTE

Engaging face of socket insert shown, cavity identification numbers are for reference only and do not appear on the part.

FIGURE 2. Insert arrangement

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Inches	mm
040	1.02
.050	1.27
100	2.54
.125	3.18
.150	3.81
2.800	71 12

NOTES

1. Dimensions are in inches
2. Metric equivalents are for general information only
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).

FIGURE 3. Layout arrangement

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REQUIREMENTS

Dimensions and configurations See figures 1 and 2

Current rating, maximum: 3 amperes per contact

Materials:

Termination organization area Potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513.

Shell. The requirements for shell materials shall be in accordance with MIL-C-83513.

Plastic body or plastic molding: Shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.

Jackpost: Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35.

Wire termination pins Wire termination pins shall conform to QQ-W-343, no. 24 AWG copper.

Mating connector: Shall conform to MIL-C-83513/1 and MIL-C-83513/3.

Plating of termination leads Solder dipping socket of termination leads will be accomplished in SN60 Pb40 or SN63 Pb37in accordance with QQ-S-571.

Part Identification Number (PIN): PIN shall consists of the letter M, the basic number of the specification sheet, a letter for the insert arrangement, a numerical code for the termination length, and a letter code for the shell finish and hardware column.

M83513/27-	H	01	C	P
Specification sheet number	Insert arrangements (see figure 2) H = 100	Termination length 01 = .109 02 = .140 03 = .172	Shell finish C = Cadmium N = Electroless nickel (space application)	Hardware N = No jackpost P = Jackpost attached

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Navy - AS, SH
Air Force - 17, 99

User activites:
Army - AT, ME
Navy - CG, MC

Preparing activity:
Air Force - 85

Agent.
DLA - ES

(Project 5935-3921-18)

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